

SIH 2026

AI-Based Legal Metrology Compliance & Inspection Platform for Packaged Commodities

SIH Problem Statement: 26034

Organization: Ministry of Consumer Affairs, Food & Public Distribution

Department: Department of Consumer Affairs (DoCA)

Category: Software

Domain: Artificial Intelligence / Computer Vision / Legal Compliance

1. Project Overview

The proposed system is an AI-assisted Legal Metrology Compliance and Inspection Platform designed to inspect packaged commodities and identify non-compliance with the Legal Metrology (Packaged Commodities) Rules, 2011.

The system allows Legal Metrology officers to capture or upload images of product packaging. AI-based image processing and OCR technologies extract important declarations such as Maximum Retail Price (MRP), Net Quantity, Manufacturer/Importer details, Consumer Care information, Dates and other mandatory declarations.

The extracted information is then passed to a Legal Metrology Rule Engine, which validates the declarations against applicable rules.

The system identifies missing or potentially incorrect declarations, highlights the suspected violations on the product image, stores evidence, and generates a structured compliance report.

The platform also provides advanced features such as risk prioritization, repeat violation detection, e-commerce listing inspection, physical package versus online listing comparison, consumer mode and inspection history.

2. Problem Understanding

Packaged commodities sold in the market are required to follow various mandatory declaration requirements under Legal Metrology regulations.

During physical inspections, officers may need to manually examine:

- Product name
- Net quantity
- MRP
- Manufacturer/Packer/Importer details
- Address
- Consumer care information
- Date-related declarations
- Other applicable mandatory declarations
- Declaration visibility and readability

Manual inspection can be:

- Time-consuming
- Difficult for large-scale inspections
- Dependent on human attention
- Difficult to maintain consistently structured evidence
- Difficult to analyze historical violations
- Difficult to compare physical packaging with online product listings

Therefore, there is a need for an AI-assisted inspection system that can reduce manual effort while keeping the final decision under human officer supervision.

3. Proposed Solution

We propose a centralized web/mobile-friendly platform where Legal Metrology officers can perform product inspections using photographs of packaged commodities.

Basic workflow:

Product → Image Capture → OCR/AI Analysis → Declaration Extraction → Rule Engine → Violation Detection → Officer Verification → Evidence → Compliance Report → Inspection History

The system does not blindly make a legal decision using AI.

Instead:

AI extracts information → Rule Engine validates applicable requirements → Officer verifies the result → System generates report.

This Human-in-the-Loop approach improves reliability and accountability.

4. Project Objectives

The main objectives of the project are:

1. Automate extraction of product declarations from packaging images.
2. Detect mandatory declarations using OCR and AI.
3. Identify missing declarations.
4. Validate declarations using a configurable Legal Metrology rule engine.
5. Detect MRP and Net Quantity information.
6. Identify Manufacturer/Packer/Importer details.
7. Detect Consumer Care information.
8. Perform basic readability and declaration-placement analysis.
9. Highlight suspected violations on product images.
10. Maintain inspection evidence.
11. Generate compliance reports.
12. Maintain product and inspection history.
13. Help officers identify repeat violations.
14. Prioritize high-risk products/manufacturers.
15. Compare physical packaging with online product listings.
16. Provide a consumer-friendly mode for product information.

5. Features

5.1 Must-Have Features

1. Product Image Scanning

Officer can upload or capture multiple images of a product from different sides.

2. OCR

OCR extracts text from product packaging images.

3. Declaration Extraction

AI identifies important declarations from extracted text.

Examples:

- Product Name
- MRP
- Net Quantity
- Manufacturer
- Packer
- Importer
- Address
- Consumer Care
- Date-related information

4. Legal Metrology Rule Engine

A configurable rule engine checks extracted information against applicable Legal Metrology requirements.

5. Missing Declaration Detection

The system detects declarations that appear to be missing.

6. MRP Detection

The system identifies MRP information and its location on the package.

7. Net Quantity Detection

The system identifies quantity declarations such as:

- g
- kg
- ml
- L
- cm
- other applicable units

8. Manufacturer/Importer Detection

The system identifies manufacturer, packer and importer information where applicable.

9. Consumer Care Detection

The system searches for consumer-care/contact information.

10. Font & Readability Analysis

The system performs basic analysis of:

- Text visibility
- Text size
- Image clarity
- Contrast
- Readability

11. Violation Highlighting

Potential violations are highlighted using bounding boxes or markers on the product image.

12. Evidence Capture

The system stores:

- Product photographs
- OCR output
- Detected violation
- Timestamp
- Inspection ID
- Officer information

13. Compliance Report

The system generates a structured compliance report.

14. Officer Dashboard

Officers can view:

- Total inspections
- Compliant products
- Non-compliant products
- Pending reviews
- Recent inspections
- High-risk products

15. Inspection History

All inspections are stored and can be searched later.

6. USP / Innovation Features

6.1 Consumer Mode

A simplified interface allows consumers to scan a product and view available information such as:

- MRP
- Net quantity
- Manufacturer
- Consumer care
- Ingredients
- Nutrition information
- Warning information

This mode is informational and does not replace official regulatory inspection.

6.2 Ingredient Extraction

The system extracts ingredient information from product labels using OCR and NLP.

6.3 Package Warning Extraction

The system identifies warnings and caution statements present on the packaging.

6.4 Nutrition Information

For applicable products, the system can extract:

- Calories
- Sugar
- Fat
- Saturated fat
- Protein
- Sodium/Salt
- Fibre

6.5 A–E Nutrition Profile

The system can provide a consumer-friendly nutrition profile from A to E, where A represents a comparatively better nutritional profile and E represents a comparatively poorer profile based on a defined methodology.

This feature is separate from Legal Metrology compliance and should not be represented as an official government rating.

6.6 Product Comparison

Users can compare products based on extracted information such as:

- **MRP**
- **Quantity**
- **Ingredients**
- **Nutrition**
- **Manufacturer**
- **Packaging declarations**

6.7 E-Commerce Listing Inspection

The system can inspect product information available on online marketplaces or e-commerce listings.

It can extract information such as:

- Product title
- MRP
- Selling price
- Net quantity
- Manufacturer
- Country of origin
- Other displayed declarations

6.8 Physical Package vs Online Listing Comparison

One of the major USP features is comparing:

Physical Product ↔ Online Product Listing

The system can identify potential mismatches such as:

- Different MRP
- Different quantity
- Different manufacturer information
- Different product name
- Missing declarations online
- Different product packaging/details

7. Advanced Features

7.1 Manufacturer Self-Audit

Manufacturers can upload their own packaging designs before releasing products.

The system can identify potentially missing declarations and provide a pre-check.

7.2 Risk Prioritization

The platform can assign risk levels to products/manufacturers.

Example:

- Low Risk
- Medium Risk
- High Risk
- Critical Risk

Risk can be calculated using factors such as:

- Previous violations
- Number of violations
- Severity
- Product category
- Repeat violations

7.3 Consumer Complaints

Consumers can submit complaints related to suspected packaging or declaration issues.

7.4 Multilingual Interface

The system can support multiple Indian languages for improved accessibility.

7.5 Offline Inspection

Officers can capture product information even with limited/no internet connectivity.

Data can synchronize with the central server when connectivity becomes available.

7.6 QR/Barcode Product Identity

The system can scan:

- QR codes
- Barcodes

to quickly identify or retrieve product records.

7.7 AI Confidence + Human Review

Every AI extraction can have a confidence score.

Example:

MRP: ₹199 — Confidence: 96%

If confidence is low, the system asks the officer to verify the information.

7.8 Audit Trail

Important activities are recorded, including:

- Who performed inspection
- When inspection was performed
- What information was detected
- What was modified
- Who approved the result

7.9 Repeat Violation Detection

The system can identify manufacturers/products with repeated violations.

8. User Roles

8.1 Legal Metrology Officer

Officer can:

- Login
- Create inspection
- Capture product images
- Run AI inspection
- Review extracted declarations
- Verify violations
- Add remarks
- Capture evidence
- Generate reports
- View inspection history

8.2 Administrator

Administrator can:

- Manage officers
- Manage users
- Configure rules
- View overall analytics
- Manage product database
- Monitor system activity
- Manage audit logs

8.3 Manufacturer

Manufacturer can:

- Perform self-audit
- Upload label designs
- Check potential violations
- View audit results

8.4 Consumer

Consumer can:

- Scan product
- View extracted information
- Compare products
- Submit complaints

9. Complete System Workflow

Step 1 – Login

Officer logs into the secure system.

Step 2 – Create New Inspection

Officer creates a new inspection.

System generates a unique Inspection ID.

Step 3 – Product Identification

Officer can enter or scan:

- Barcode
- QR code
- Product ID

Step 4 – Image Capture

Officer captures multiple images:

- Front
- Back
- Left
- Right
- Top
- Bottom

Step 5 – Image Processing

The system performs:

- Image enhancement
- Cropping
- Perspective correction
- Noise removal

Step 6 – OCR

OCR extracts text from the product images.

Step 7 – AI/NLP Processing

The AI categorizes extracted text into declarations.

Example:

MRP ₹199

→ MRP

Net Qty 500 g

→ Net Quantity

Manufactured by ABC Foods

→ Manufacturer

Step 8 – Rule Engine

The extracted information is checked against the applicable Legal Metrology rules.

Step 9 – Violation Detection

Potential issues are classified.

Example:

Missing Declaration → Consumer Care Information

Step 10 – Image Highlighting

The system highlights the relevant area of the package.

Step 11 – Human Verification

Officer reviews the AI result.

Officer can:

- Accept
- Reject
- Edit
- Add remarks

Step 12 – Compliance Result

System produces:

- Compliant
- Potentially Non-Compliant
- Requires Officer Review

Step 13 – Evidence Storage

Images and inspection information are stored securely.

Step 14 – Report Generation

System generates PDF/editable compliance report.

Step 15 – Inspection History

Inspection is stored for future reference.

10. AI/ML Components

The AI layer can contain the following components:

Computer Vision

Used for:

- Product image processing
- Text-region detection
- Declaration-region detection
- Violation highlighting

OCR

Possible technologies:

- Tesseract OCR
- EasyOCR
- PaddleOCR

NLP

Used for:

- Text classification
- Declaration extraction
- Entity recognition
- Manufacturer extraction
- Consumer-care extraction

Object Detection

Models such as YOLO can be used to detect specific regions/declaration areas.

Confidence Scoring

Each AI prediction can include a confidence score.

11. AI vs Rule Engine

This is a very important architecture principle.

AI System	Rule Engine
Reads text	Applies legal rules
Extracts declarations	Validates declarations
Detects MRP	Checks MRP compliance
Detects quantity	Checks quantity requirements
Detects manufacturer	Checks manufacturer declaration
Detects text regions	Checks placement/readability
Provides confidence	Produces rule-based compliance result
Highlights suspected issues	Classifies compliance status

Final Principle

AI should assist the officer, not independently make the final legal decision.

The final regulatory determination remains subject to applicable law, rules, amendments, and authorized officer verification.

12. Legal Metrology Rule Engine

The rule engine is the core compliance component.

It should be designed as a configurable and version-controlled rules system.

Each rule can contain:

- Rule ID
- Rule description
- Applicable product category
- Required declaration
- Validation condition
- Severity
- Effective date/version
- Evidence requirement

Example:

Rule: Net Quantity Declaration

Input: OCR extracted quantity

Validation: Check whether required quantity information is present and appropriately declared according to the applicable rule.

Result:

- Pass
- Fail
- Review Required

The rules should be updated when Legal Metrology regulations/amendments change.

13. Compliance Scoring

The system can generate an internal compliance score to help officers prioritize reviews.

Example:

Compliance Score: 82/100

Possible categories:

- Mandatory declarations
- MRP
- Net quantity
- Manufacturer/importer information
- Consumer care
- Readability
- Other applicable checks

However, the score should be treated as a decision-support indicator, not as a replacement for the legally applicable compliance determination.

14. Violation Classification

Potential violations can be classified as:

Critical

Potentially serious compliance issue requiring immediate officer attention.

Major

Important declaration/compliance issue.

Minor

Less severe formatting/readability issue.

Review Required

AI is uncertain and human verification is required.

15. Evidence Management

For every inspection, the system should maintain:

- Inspection ID
- Product ID
- Product images
- OCR output
- Detected declarations
- Violation location
- Violation description
- Timestamp
- Officer details
- Officer remarks
- Verification status
- Report

Evidence should be stored securely and should not be silently modified.

16. Inspection History

The system maintains historical inspection records.

Officers can search by:

- Product
- Manufacturer
- Date
- Inspection ID
- Violation
- Risk level
- Compliance status

This enables identification of:

First Violation → Previous Violation → Repeat Violation

17. Risk Prioritization

The platform can generate risk profiles.

Example:

Manufacturer A

- Previous inspections: 20
- Violations: 8
- Repeat violations: 5
- Risk level: High

The system can help officers prioritize inspections based on historical patterns.

18. System Architecture

The proposed architecture consists of:

Frontend Layer

- Officer Dashboard
- Admin Dashboard
- Manufacturer Portal
- Consumer Mode

Backend Layer

- Authentication
- Inspection Management
- Product Management
- Report Generation
- Rule Engine
- Evidence Management

AI Layer

- Image Processing
- OCR
- Computer Vision
- NLP
- Declaration Extraction
- Confidence Scoring

Database Layer

Stores:

- Users
- Products
- Inspections
- Declarations
- Violations
- Evidence
- Rules
- Reports
- Audit Logs

External/Integration Layer

- E-commerce listings
- Barcode/QR systems
- Government rule/document sources where permitted

19. Technology Stack

Frontend

- HTML
- CSS
- JavaScript
- React.js

Backend

Primary option:

- REST APIs

Alternative AI service:

- Python
- Flask/FastAPI

AI/ML

- Python
- OpenCV
- YOLO

- Tesseract/EasyOCR/PaddleOCR
- NLP
- NumPy
- Pandas
- scikit-learn

Database

- Firebase

Authentication

- JWT
- Role-Based Access Control

Storage

- Cloud object storage/local secure storage depending on deployment requirements.

Deployment

-
- Easy Way Vercel

20. Database Design (Firebase)

Major tables/collections:

Users

- User ID
- Name
- Email
- Role
- Password Hash
- Status

Products

- Product ID
- Product Name
- Manufacturer
- Category

- Barcode
- QR Code
- Created Date

Inspections

- Inspection ID
- Product ID
- Officer ID
- Date
- Status
- Risk Level
- Compliance Result

Declarations

- Declaration ID
- Inspection ID
- Declaration Type
- Extracted Value
- Confidence
- Verification Status

Violations

- Violation ID
- Inspection ID
- Rule ID
- Violation Type
- Severity
- Description
- Status

Evidence

- Evidence ID
- Inspection ID
- Image
- Timestamp
- Bounding Box
- Hash/Integrity Metadata

Rules

- Rule ID
- Rule Version
- Rule Description
- Category
- Validation Logic
- Effective Date

Audit Logs

- Log ID
- User ID
- Action
- Timestamp
- Record ID

21. Security & Privacy

The platform should implement:

- Secure authentication
- JWT/session security
- Role-based access
- Password hashing
- HTTPS
- Input validation
- API authorization
- Secure file upload
- File type validation
- Audit logging
- Database access control
- Backup and recovery

Sensitive information should not be unnecessarily exposed.

22. AI & Legal Precautions

AI results can contain errors due to:

- Poor image quality
- Reflections
- Curved packaging
- Small text
- Damaged packaging
- Multiple languages
- Stylized fonts

Therefore, the platform should provide:

AI Detection → Confidence → Human Review → Final Action

The system should clearly distinguish between:

AI-generated suggestion and legally verified decision.

23. UI/UX Screens

The proposed system can contain the following screens:

1. Login
2. Officer Dashboard
3. New Inspection
4. Product Identification
5. Image Capture/Upload
6. AI Scanning
7. OCR Results
8. Declaration Review
9. Compliance Checklist
10. Violation Details
11. Evidence Viewer
12. Compliance Report
13. Inspection History
14. Product Repository
15. Risk Dashboard
16. Manufacturer Analytics
17. Admin Dashboard
18. Consumer Mode

19. Product Comparison

20. E-commerce Comparison

24. Report Format

The generated report should contain:

Inspection Information

- Inspection ID
- Date & Time
- Officer
- Product ID

Product Information

- Product Name
- Manufacturer
- Category
- Barcode/QR

Extracted Declarations

- MRP
- Net Quantity
- Manufacturer
- Importer
- Consumer Care
- Other declarations

Compliance Analysis

- Rule checked
- Result
- Confidence
- Officer verification

Violations

- Violation
- Severity
- Applicable rule

- Evidence
- Officer remarks

Final Result

- Compliant
- Non-Compliant
- Review Required

Evidence

Product images with highlighted areas.

25. Offline Mode

For field inspections where internet connectivity is unavailable:

1. Officer opens inspection application.
2. Captures product images.
3. Basic processing and data storage happens locally.
4. Inspection remains in pending synchronization state.
5. When internet becomes available, data is synchronized with the central server.

The system should clearly show:

Synced / Pending Sync / Sync Failed

26. Testing Strategy

Testing will include:

Unit Testing

Individual modules are tested separately.

Integration Testing

AI, backend, database and rule engine are tested together.

OCR Testing

Measure accuracy of extracted text.

AI Testing

Evaluate:

- MRP detection
- Quantity detection
- Manufacturer detection

- Declaration detection

Rule Engine Testing

Test multiple compliant and non-compliant cases.

Security Testing

Test:

- Authentication
- Authorization
- File upload
- API security
- Injection attacks

User Acceptance Testing

Legal/compliance workflow should be reviewed with domain experts where possible.

27. Evaluation Metrics

Important metrics include:

OCR Accuracy

Percentage of correctly extracted text.

Declaration Extraction Accuracy

Percentage of correctly identified declarations.

MRP Detection Accuracy

Correct detection of MRP values.

Net Quantity Detection Accuracy

Correct extraction of quantity and unit.

Violation Detection Precision

Percentage of detected violations that are actually valid issues.

Violation Detection Recall

Percentage of actual issues successfully detected.

Processing Time

Average time required to analyze one product.

Human Review Time

Reduction in officer's manual inspection effort.

28. Deployment Plan

The system can be deployed as a cloud-based application.

Development

Local development environment.

Testing

Staging environment.

Production

Secure cloud infrastructure.

Possible deployment architecture:

Frontend → API Gateway → Spring Boot Backend → Python AI Service → Database + File Storage

Docker can be used for containerization.

29. Development Roadmap

Phase 1 – Basic Platform

- Login
- Officer dashboard
- Product management
- Inspection creation
- Image upload

Phase 2 – AI

- Image preprocessing
- OCR
- Declaration extraction
- MRP detection
- Quantity detection

Phase 3 – Rule Engine

- Rule database
- Compliance checks
- Missing declaration detection
- Violation classification

Phase 4 – Evidence & Reports

- Image highlighting
- Evidence storage
- PDF report
- Inspection history

Phase 5 – Advanced Features

- Risk prioritization
- Repeat violation detection
- E-commerce comparison
- Consumer mode
- Product comparison

Phase 6 – Optimization

- Multilingual support
- Offline mode
- Performance optimization
- Security testing
- Deployment

30. USP / Innovation

The primary innovation of the project is not simply OCR or image scanning.

The platform combines:

AI + Computer Vision + OCR + Legal Rule Engine + Human Verification + Evidence Management + Risk Analytics + E-commerce Cross-Validation into one integrated inspection platform.

Three strongest USPs

1. Physical Package ↔ Online Listing Mismatch

Detect potential differences between the physical product and its online listing.

2. Risk-Based Enforcement

Use historical inspection data to identify repeat offenders and high-risk products.

3. AI + Evidence + Human Verification

AI assists the officer while maintaining human verification, evidence capture and audit trail.

31. Existing Solution Gap / Competitive Differentiation

Existing label-compliance tools already provide some capabilities such as label checking, declaration validation and packaging workflow.

Our project does not claim that existing tools are incapable.

Instead, our differentiation is the integration of:

Capability	Existing Label Tools	Proposed Platform
OCR	Available in some	✓
Declaration extraction	Available in some	✓
MRP detection	Available in some	✓
Rule validation	Available in some	✓
Field officer workflow	Limited/varies	✓
Evidence management	Limited/varies	✓
Inspection history	Limited/varies	✓
Risk prioritization	Limited/varies	✓
Repeat violation detection	Limited/varies	✓
Consumer mode	Limited/varies	✓
Physical vs online comparison	Limited/varies	✓
E-commerce inspection	Limited/varies	✓
Human-in-the-loop review	Varies	✓
Audit trail	Varies	✓
Offline inspection	Limited/varies	✓

Positioning

“An AI-assisted field inspection and enforcement intelligence platform for Legal Metrology officers.”

32. Expected Benefits

For Legal Metrology Officers

- Faster inspection
- Reduced manual effort
- Structured evidence
- Faster report generation
- Historical inspection access
- Risk-based prioritization

For Government

- Better enforcement intelligence
- Centralized inspection data
- Repeat violation tracking
- Analytics

- Standardized inspection workflow

For Manufacturers

- Pre-compliance checking
- Self-audit
- Reduced packaging errors

For Consumers

- Better access to product information
- Product comparison
- Complaint submission
- Improved awareness

33. Future Scope

Future versions can include:

- Advanced multilingual OCR
- On-device AI
- More advanced computer vision
- Voice-assisted inspection
- Mobile application
- Automated product recognition
- Advanced fraud/mismatch detection
- Large-scale government deployment
- Integration with authorized government systems
- Advanced analytics and dashboards
- Continuous rule updates
- AI-assisted inspection prioritization

34. Final Project Flow

Officer Login



Create Inspection



Scan Product / Barcode / QR



Capture Product Images



Image Preprocessing



OCR



AI Declaration Extraction



MRP / Quantity / Manufacturer / Consumer Care Detection



Legal Metrology Rule Engine



Compliance Analysis



Violation Detection



Violation Highlighting



AI Confidence Score



Human Officer Verification



Evidence Capture



Compliance Result



PDF / Editable Report



Inspection History



Risk & Analytics

35. Conclusion

The proposed AI-Based Legal Metrology Compliance & Inspection Platform provides an integrated approach for inspecting packaged commodities.

Instead of relying completely on manual inspection, the platform uses AI, OCR and computer vision to assist officers in extracting and analyzing product declarations.

The Legal Metrology Rule Engine provides structured rule-based validation, while Human-in-the-Loop verification ensures that AI output is reviewed before being used for official action.

The combination of field inspection, evidence management, inspection history, risk prioritization, repeat violation detection and physical-versus-online product comparison makes the proposed system more than a simple label scanner.

The ultimate goal is to create a scalable, secure and intelligent decision-support system for modern Legal Metrology inspection and enforcement.